# CIS 3353 - Group Project Presentation

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# Project Overview

Rationale:

This project focuses on demonstrating how a network can be vulnerable when basic security practices are not implemented. Many systems rely on default or insecure configurations, which can expose sensitive data and services to attacks. By building and securing this environment step-by-step, the project highlights the importance of proper network configuration, encryption, and access control. This approach benefits students and future IT/security professionals by providing hands-on experience with real-world security concepts.

Context:

In today's environment, organizations commonly use tools such as firewalls, encryption (HTTPS), and secure authentication methods to protect their systems. However, without proper configuration, these tools may be ineffective or misused. This project simulates a real-world scenario by starting with an insecure setup (HTTP, weak access controls) and progressively applying security measures such as TLS encryption, SSH key-based authentication, and network segmentation. It helps bridge the gap between theoretical knowledge and practical implementation.

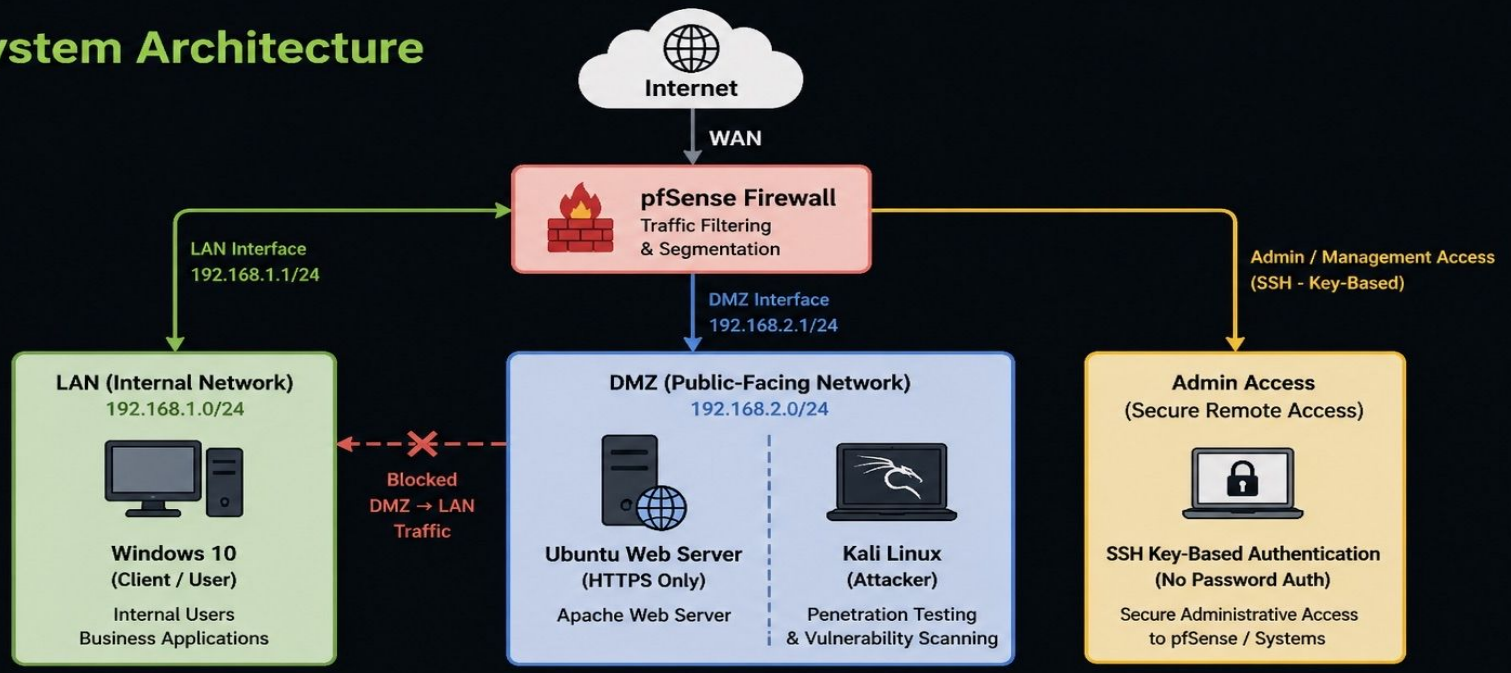
# Sprint Overview

Work was completed across three main sprints, with the majority of technical implementation and configuration handled by Nastassja.

- Sprint 1: Focused on defining roles, resolving initial team coordination challenges, and setting up the virtual machine environment required for development.
- Sprint 2: Consisted of completing the bulk of user stories and technical tasks, including network deployment, vulnerability testing, and establishing the GitHub Wiki for documentation.
- Sprint 3: Focused on finalizing remaining issues, validating security improvements, organizing project materials, and completing presentation deliverables.

While GitHub reflects a higher volume of commits from one contributor, additional work such as documentation support and validation was performed collaboratively.

# System Architecture




## Legend

- LAN Traffic (Allowed)
- DMZ Traffic (Allowed)
- Admin Access (Allowed)
- Blocked Traffic

## Traffic Rules (pfSense Firewall)

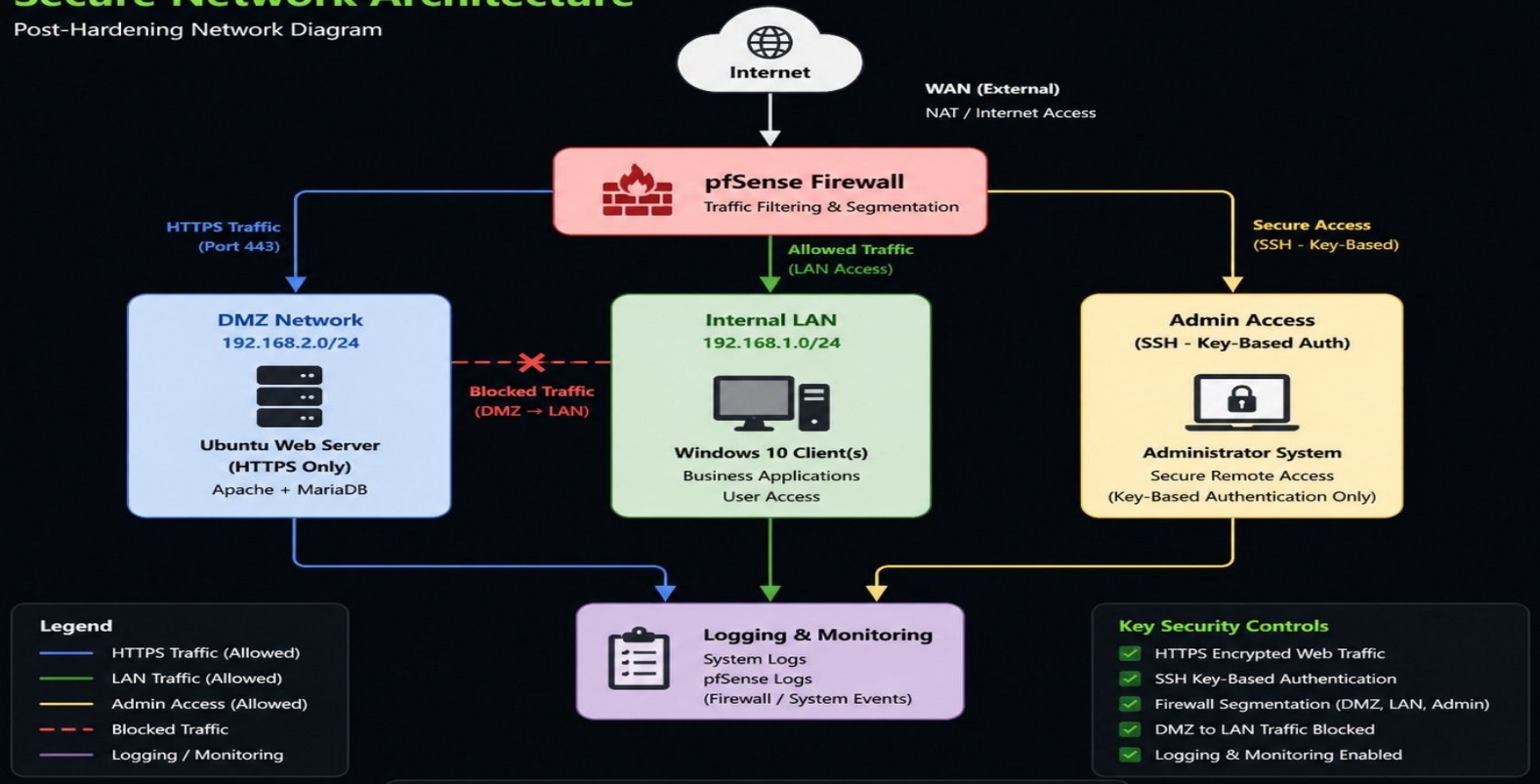
- ✅ Internet → DMZ (HTTPS 443 Allowed)
- ✅ Internet → LAN (Allowed as needed)
- ✅ Admin → pfSense / Systems (SSH Key-Based Only)
- ❌ DMZ → LAN (Blocked)
- ❌ Unnecessary Ports/Services (Blocked)

## Security Features Implemented

- ✅ HTTPS Encryption (TLS/SSL)
- ✅ SSH Key-Based Authentication
- ✅ Network Segmentation (LAN / DMZ)
- ✅ Firewall Rules & Access Control
- ✅ Vulnerability Scanning & Monitoring

# Secure Network Architecture

Post-Hardening Network Diagram



# Our Objective

This project demonstrates the process of building a vulnerable network environment and progressively securing it using industry-standard cybersecurity practices. The system begins in an intentionally insecure state, exposing services such as HTTP, and is then hardened through the implementation of HTTPS (TLS), SSH key-based authentication, and firewall segmentation.

# Milestones

Milestone 1 - Project Setup & Architecture Design

- Defined project scope and structure
- Designed network architecture and IP scheme
- Established GitHub Wiki for documentation

Milestone 2 - Base Infrastructure Deployment

- Deployed pfSense and configured network segmentation
- Set up Ubuntu Server in a insecure state
- Verified connectivity across WAN, LAN, DMZ

Milestone 3 - Threat Demonstration

- Installed Kali Linux for testing
- Performed reconnaissance and vulnerability scanning
- Captured unencrypted HTTP traffic
- Documented identified weaknesses

Milestone 4 - Security Implementation

- Enabled HTTPS using TLS encryption
- Enforced SSH key-based authentication
- Implemented pfSense firewall rules
- Hardened system configurations

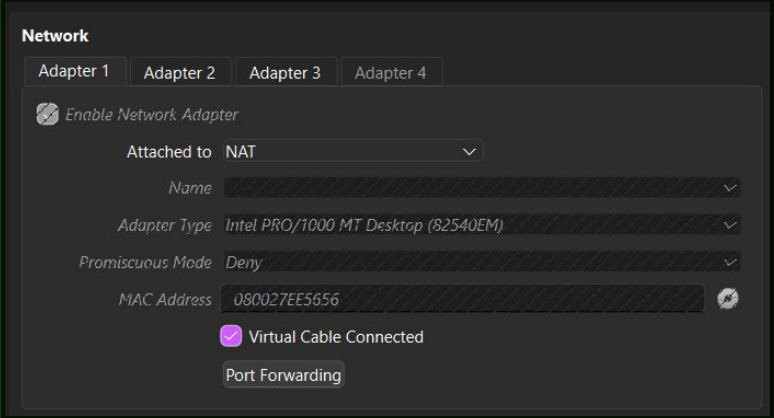
Milestone 5 - Verification & Documentation

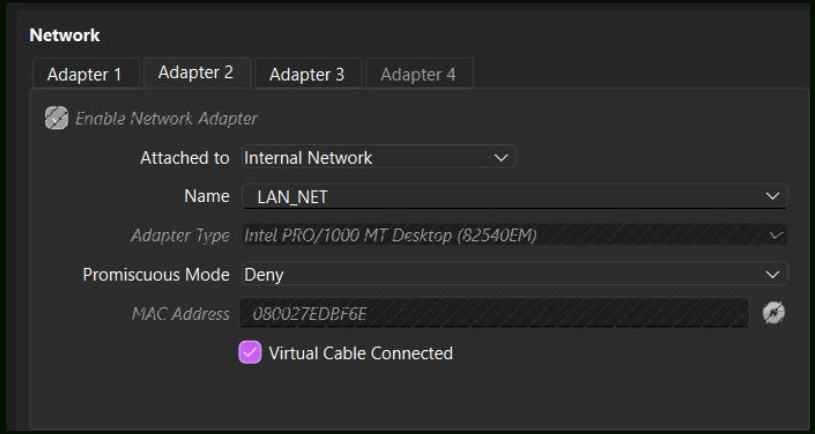
- Re-ran vulnerability scans to validate fixes
- Captured encrypted traffic for comparison
- Demonstrated effectiveness of security controls
- Finalized documentation and presentation

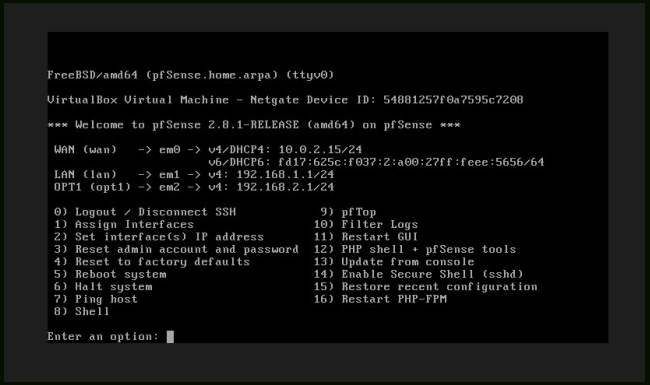
# Milestone 1 - Project Setup & Architecture Design

| User Story # | Title/Description | Assigned To | Committed By | Status |
|---|---|---|---|---|
| US-03 | As a team, we want to structure the GitHub Wiki for documentation | Nastassja | Nastassja | Complete |

# M1 - Evidence



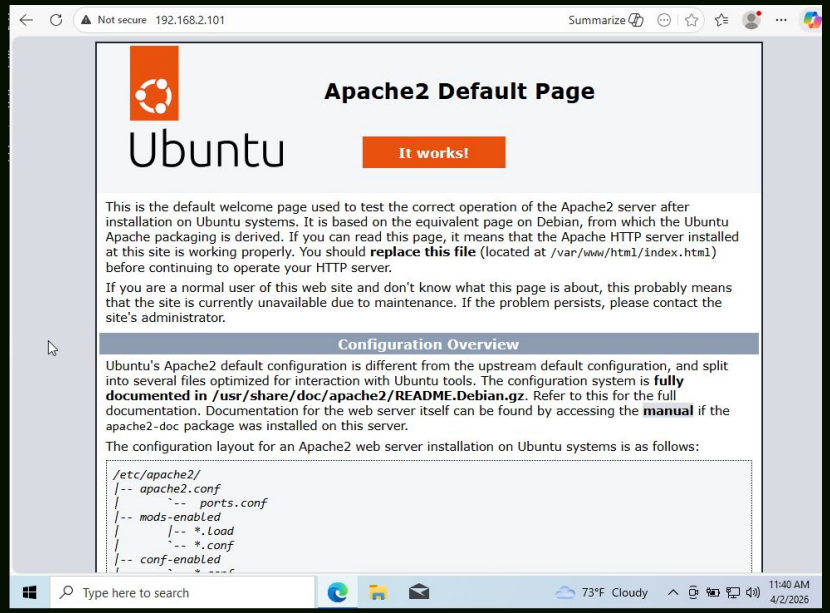



```
FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 54881257f0a7595c7200

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan)   -> em0 -> v4/DHCP4: 10.0.2.15/24
                       v6/DHCP6: fd17:625c:f037:2:a00:27ff:feee:5656/64
 LAN (lan)   -> em1 -> v4: 192.168.1.1/24
 OPT1 (opt1) -> em2 -> v4: 192.168.2.1/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```

# Milestone 2 - Base Infrastructure Deployment

| User Story # | Title/Description | Assigned To | Committed By | Status |
|---|---|---|---|---|
| US-01 | As a network admin, I want to deploy pfSense to segment the network | Nastassja | Nastassja | Complete |
| US-02 | As a system admin, I want to deploy an insecure Ubuntu Server for testing | Nastassja | Nastassja | Complete |

# M2 - Evidence




```
Last login: Thu Apr  2 16:10:22 UTC 2026 on tty1
vboxuser@Ubuntu-DMZ:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=1.54 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=0.953 ms
64 bytes from 192.168.2.1: icmp_seq=3 ttl=64 time=0.863 ms
64 bytes from 192.168.2.1: icmp_seq=4 ttl=64 time=1.10 ms
64 bytes from 192.168.2.1: icmp_seq=5 ttl=64 time=1.12 ms
64 bytes from 192.168.2.1: icmp_seq=6 ttl=64 time=1.08 ms
64 bytes from 192.168.2.1: icmp_seq=7 ttl=64 time=1.06 ms
64 bytes from 192.168.2.1: icmp_seq=8 ttl=64 time=0.929 ms
64 bytes from 192.168.2.1: icmp_seq=9 ttl=64 time=1.43 ms
64 bytes from 192.168.2.1: icmp_seq=10 ttl=64 time=1.14 ms
^C
--- 192.168.2.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 0.863/1.120/1.544/0.203 ms
vboxuser@Ubuntu-DMZ:~$ _
```

```
 1) Assign Interfaces                10) Filter Logs
 2) Set interface(s) IP address      11) Restart GUI
 3) Reset admin account and password 12) PHP shell + pfSense tools
 4) Reset to factory defaults        13) Update from console
 5) Reboot system                    14) Enable Secure Shell (sshd)
 6) Halt system                      15) Restore recent configuration
 7) Ping host                        16) Restart PHP-FPM
 8) Shell

Enter an option: 7


Enter a host name or IP address: 192.168.2.1

PING 192.168.2.1 (192.168.2.1): 56 data bytes
64 bytes from 192.168.2.1: icmp_seq=0 ttl=64 time=3.827 ms
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=0.240 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=0.173 ms

--- 192.168.2.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.173/1.413/3.827/1.707 ms

Press ENTER to continue.
```

# Milestone 3 - Threat Demonstration (Insecure State)

| User Story # | Title/Description | Assigned To | Committed By | Status |
|---|---|---|---|---|
| US-04 | As a penetration tester, I want to run reconnaissance scans on the server | Nastassja | Nastassja | Complete |
| US-05 | As a security analyst, I want to capture unencrypted HTTP traffic to show risks | Nastassja | Nastassja | Complete |

# M3 - Evidence

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 80

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000000 | 192.168.1.100 | 192.168.2.101 | TCP | 66 | 49 |
| 2 | 0.021050395 | 192.168.2.101 | 192.168.1.100 | TCP | 66 | 80 |
| 3 | 0.021051070 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 49 |
| 4 | 0.021051182 | 192.168.1.100 | 192.168.2.101 | TCP | 66 | 49 |
| 5 | 0.021051253 | 192.168.2.101 | 192.168.1.100 | TCP | 66 | 80 |
| 6 | 0.021051345 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 49 |
| 7 | 0.064549813 | 192.168.1.100 | 192.168.2.101 | HTTP | 614 | GE |
| 8 | 0.064550646 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | 80 |
| 9 | 0.064550729 | 192.168.2.101 | 192.168.1.100 | TCP | 2974 | 80 |
| 10 | 0.064550768 | 192.168.2.101 | 192.168.1.100 | HTTP | 594 | HT |
| 11 | 0.068580622 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 49 |

```
▸ Frame 1: Packet, 66 bytes on wire (528
▸ Ethernet II, Src: PCSSystemtec_f2:d0:9a
▸ Internet Protocol Version 4, Src: 192.1
▸ Transmission Control Protocol, Src Port

0000  08 00 27 55 95 e5 08 00  27 f2 d0
0010  00 34 99 34 40 00 7f 06  dd 75 c0
0020  02 65 c2 60 00 50 a8 3e  1b f7 00
0030  fa f0 68 1f 00 00 02 04  05 b4 01
0040  04 02
```

wireshark_eth08GKYN3.pcapng | Packets: 51 · Displayed: 46 (90.2%) | Profile: Default

Wireshark · Follow TCP Stream (tcp.stream eq 3) · eth0

```
GET / HTTP/1.1
Host: 192.168.2.101
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0
.0.0 Safari/537.36 Edg/146.0.0.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,
application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9


HTTP/1.1 200 OK
Date: Mon, 20 Apr 2026 02:09:50 GMT
Server: Apache/2.4.52 (Ubuntu)
Last-Modified: Thu, 02 Apr 2026 15:25:56 GMT
ETag: "29af-64e7bd12268ed-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 3121
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html
```

kali@kali: ~

Session Actions Edit View Help

```
s)
                to their respective packages, not to the web server itself.
          </p>
        </div>

      </div>
    </div>
    <div class="validator">
    </div>
  </body>
</html>

┌──(kali㉿kali)-[~]
└─$ sudo nmap 192.168.2.101
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-08 17:37 EDT
Nmap scan report for 192.168.2.101
Host is up (0.00085s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 08:00:27:55:95:E5 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 1.08 seconds

┌──(kali㉿kali)-[~]
└─$
```

# Milestone 4 - Security Implementation and Cryptography

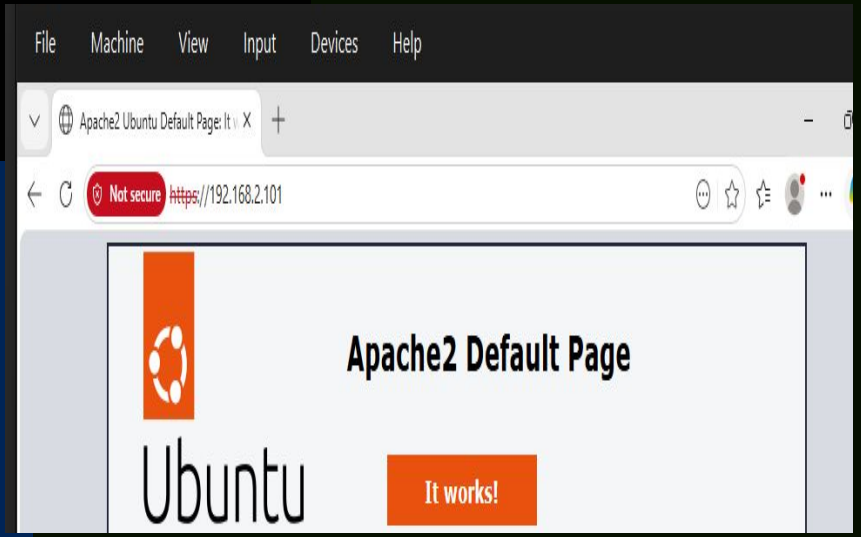
| User Story # | Title/Description | Assigned To | Committed By | Status |
|---|---|---|---|---|
| US-06 | As a system admin, I want to implement TLS to secure the web server | Nastassja | Nastassja | Complete |
| US-07 | As a system admin, I want to enforce SSH key authentication | Nastassja | Nastassja | Complete |
| US-08 | As a network admin, I want to enforce firewall rules for segmentation | Nastassja | Nastassja | Complete |

# M4 - Evidence

File Machine View Input Devices Help

```
vboxuser@Ubuntu-DMZ:~$ ping 192.168.1.100
PING 192.168.1.100 (192.168.1.100) 56(84) bytes of data.
^C
--- 192.168.1.100 ping statistics ---
23 packets transmitted, 0 received, 100% packet loss, time 22559ms

vboxuser@Ubuntu-DMZ:~$
```

```
PS C:\Users\vboxuser> Test-NetConnection 192.168.2.101 -Port 22
WARNING: TCP connect to (192.168.2.101 : 22) failed
WARNING: Ping to 192.168.2.101 failed with status: TimedOut


ComputerName            : 192.168.2.101
RemoteAddress           : 192.168.2.101
RemotePort              : 22
InterfaceAlias          : Ethernet 2
SourceAddress           : 192.168.1.100
PingSucceeded           : False
PingReplyDetails (RTT)  : 0 ms
TcpTestSucceeded        : False


PS C:\Users\vboxuser>
```



# Milestone 5 - Verification and Documentation

| User Story # | Title/Description | Assigned To | Committed By | Status |
|---|---|---|---|---|
| US-09 | As a security analyst, I want to re-run scans to verify fixes | Nastassja | Nastassja | Complete |
| US-10 | As a team, we want to finalize documentation and presentation | Nastassja, Nickolas, Shaleigh | Nastassja, Nickolas, Shaleigh | Complete |

# M5 - Evidence

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 443

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 112 | 15.005962493 | 192.168.1.100 | 192.168.2.101 | TLSv1.3 | 84 | Change Cipher Spec, Application Data |
| 113 | 15.007850891 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 50266 → 443 [FIN, ACK] Seq=1732 Ack=1601 Wi |
| 114 | 15.007851207 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | 443 → 50266 [FIN, ACK] Seq=1601 Ack=1732 Wi |
| 115 | 15.007851229 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | 443 → 50266 [ACK] Seq=1602 Ack=1733 Win=641 |
| 116 | 15.134882438 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 50263 → 443 [ACK] Seq=5275 Ack=19697 Win=26 |
| 117 | 15.224224730 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | [TCP Retransmission] 443 → 50266 [FIN, ACK] |
| 118 | 15.225719306 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | [TCP ZeroWindow] 50266 → 443 [ACK] Seq=1733 |
| 119 | 20.165133147 | 192.168.2.101 | 192.168.1.100 | TLSv1.3 | 78 | Application Data |
| 120 | 20.165329805 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | 443 → 50263 [FIN, ACK] Seq=19721 Ack=5275 W |
| 121 | 20.166404397 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | 50263 → 443 [ACK] Seq=5275 Ack=19722 Win=26 |
| 122 | 65.199866762 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | [TCP Keep-Alive] 50263 → 443 [ACK] Seq=5274 |
| 123 | 65.200087893 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | [TCP Keep-Alive ACK] 443 → 50263 [ACK] Seq= |
| 126 | 119.686656719 | 192.168.1.100 | 192.168.2.101 | TCP | 60 | [TCP Keep-Alive] 50263 → 443 [ACK] Seq=5274 |
| 127 | 119.687252865 | 192.168.2.101 | 192.168.1.100 | TCP | 60 | 443 → 50263 [RST] Seq=19722 Win=0 Len=0 |

```
▸ Frame 117: Packet, 60 bytes on wire (480 bits), 60 bytes ca
▸ Ethernet II, Src: PCSSystemtec_55:95:e5 (08:00:27:55:95:e5)
▸ Internet Protocol Version 4, Src: 192.168.2.101, Dst: 192.1
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 502

0000  08 00 27 f2 d0 9a 08 00  27 55 95 e5 08 00 45 00   ··'
0010  00 28 fc 03 40 00 40 06  b9 b2 c0 a8 02 65 c0 a8   ·(
0020  01 64 01 bb c4 5a 7e 55  75 cd df ad b1 32 50 11   ·d
0030  01 f5 dd ab 00 00 00 00  00 00 00 00               ···
```

wireshark_eth0L5JBO3.pcapng | Packets: 127 · Displayed: 121 (95.3%) | Profile: Default

kali@kali: ~

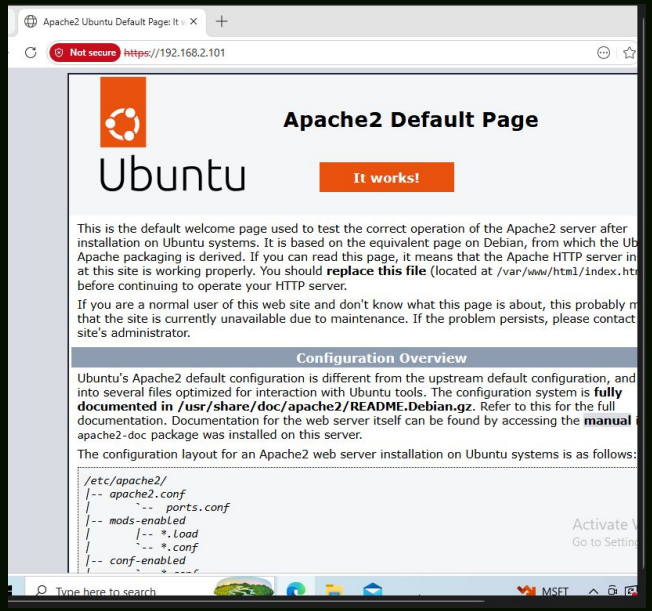
Session Actions Edit View Help

```
s)
                    to their respective packages, not to the web server itself.
            </p>
          </div>

        </div>
      </div>
      <div class="validator">
      </div>
    </body>
</html>

┌──(kali㉿kali)-[~]
└─$ sudo nmap 192.168.2.101
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-08 17:37 EDT
Nmap scan report for 192.168.2.101
Host is up (0.00085s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 08:00:27:55:95:E5 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 1.08 seconds

┌──(kali㉿kali)-[~]
└─$
```

```
└─$ nmap -sV 192.168.2.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-23 11:41 EDT
Nmap scan report for 192.168.2.101
Host is up (0.00082s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE  VERSION
22/tcp  open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.14 (Ubuntu Linux; protocol 2.0)
80/tcp  open  http     Apache httpd 2.4.52
443/tcp open  ssl/http Apache httpd 2.4.52 ((Ubuntu))
MAC Address: 08:00:27:55:95:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: 192.168.2.101; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 12.67 seconds

┌──(kali㉿kali)-[~]
└─$
```

# Prototype & Demo

This project demonstrates the creation of a segmented network environment using pfSense, an Ubuntu web server, and multiple virtual machines. The system was intentionally deployed in an insecure state and then progressively secured using HTTPS encryption, SSH key-based authentication, and firewall rules. Attacks such as port scanning and packet capture were performed to highlight vulnerabilities. The final configuration successfully prevents unauthorized access and protects data through encryption and network segmentation.




Note: We were able to implement HTTPS but still trying to input fully secure HTTPS

Project Repository: https://github.com/n4stam0on/CIS3353_Secure_Small_Business_Infrastructure

# Challenges & Limitations

Challenges:

- Uneven team contributions and lack of coordination
- Inconsistent collaboration and GitHub workflow challenges
- Risk of losing progress due to commit/branch mistakes
- Time lost troubleshooting VM setup and network issues
- Difficulty configuring pfSense (WAN/LAN/DMZ) correctly
- Connectivity issues between virtual machines
- Learning curve with tools like Kali, Wireshark, and pfSense

Limitations:

- Limited time to fully test and refine configurations
- Small lab environment (not fully reflective of real-world networks)
- Limited ability to simulate advanced or large-scale attacks
- Dependence on local VM performance and system resources
- Some tasks completed through trial-and-error rather than optimized methods

# Future Work

- Expand the network environment to include additional systems (Ex. database, Active Directory)
- Implement advanced security tools such as IDS/IPS (Snort and Suricata)
- Apply centralized logging and monitoring (SIEM Integration)
- Strengthen access control with role-based permissions and MFA
- Perform more advanced penetration testing and attack simulations
- Automate vulnerability scanning and patch management
- Improve documentation with more detailed configurations and diagrams

# Conclusion

This project has successfully demonstrated how a vulnerable network environment can be built, analyzed, and secured using industry-standard cybersecurity practices. We began by deploying a segmented network using pfSense and virtual machines, then identified key vulnerabilities such as exposed HTTP traffic and weak access controls. Through the implementation of HTTPS encryption, SSH key-based authentication, and firewall segmentation, we were able to significantly reduce these risks. The effectiveness of these controls was validated using tools like Nmap and Wireshark, confirming improvements in both security and network behavior.

Overall, this project reflects the full security lifecycle; from initial setup and vulnerability discovery to implementation and validation, highlighting the importance of proper configuration, encryption, and layered defense in protecting network environments.

Project Repository: https://github.com/n4stam0on/CIS3353_Secure_Small_Business_Infrastructure